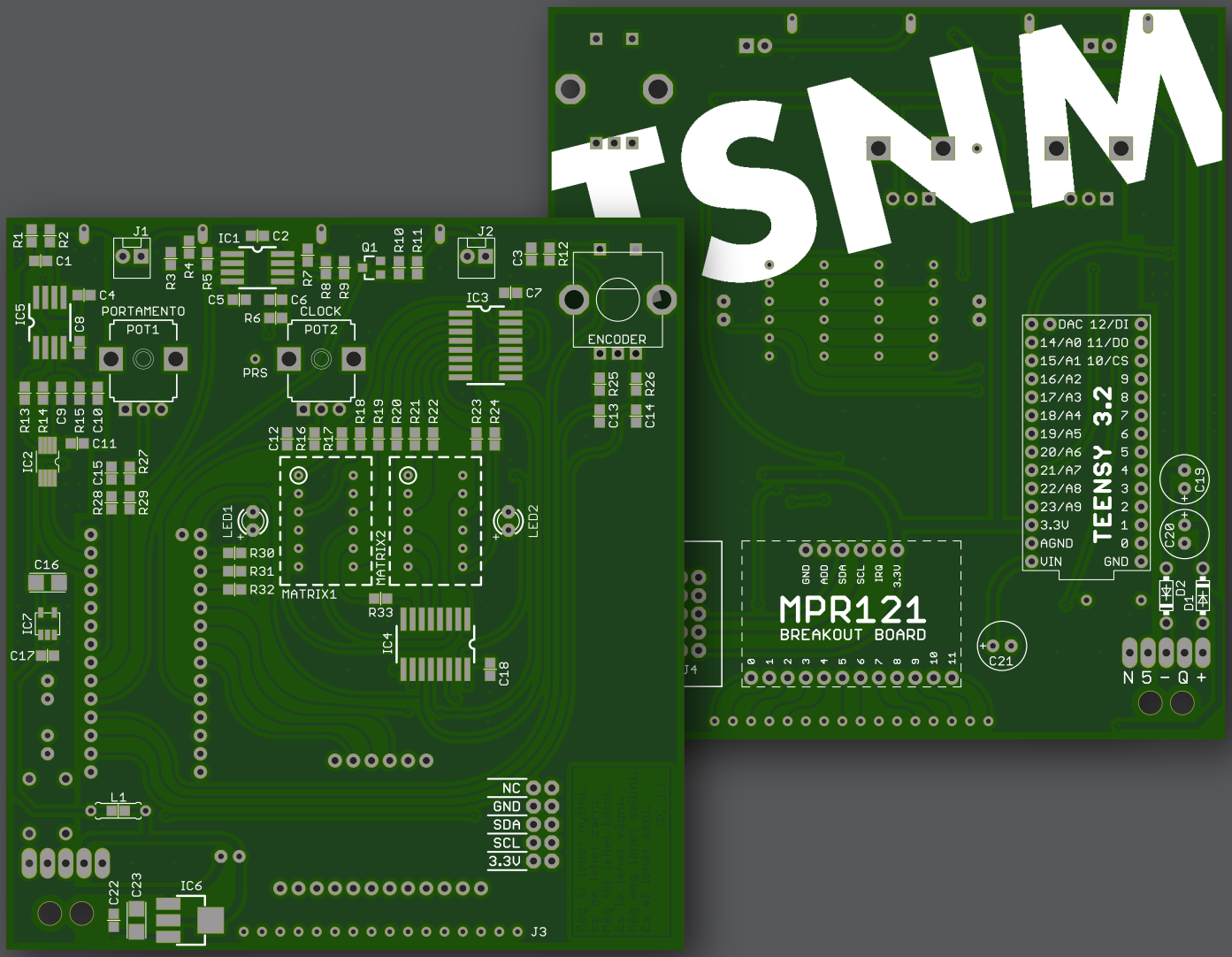


TSNM 4U
ASSEMBLY NOTES

THE INSTRUCIONS BELOVE SERVES AS A ROUGH GUIDE TO BUILD YOUR MODULE. IF YOU ARE A CONFIDENT BUILDER FEEL FREE TO FOLLOW YOUR USUAL METHODS.

SEMICONDUCTORS

Orientation of IC1-IC5 and IC7, Q1, D1 and D2 is clearly designated on the PCB. The most frightening part may be the DAC8560 (IC3), it's a VSSOP-8 device. Search YouTube for "smd ic soldering", then prepare your flux and solder wick.

RESISTORS, CAPACITORS, FILTER BEADS

First of all solder resistors with 0.1% tolerance (R1, R2, R13, R14, R15), then the rest. After finishing the module you may want to change R30, R31. They act as current limiting resistors, brightness of the LEDs (LED1, LED2) depends of their value.

HEADERS, SOCKETS, JACKS, TH CAPS

Solder the sockets for Teensy and MPR121 boards, expander connector (EXP) and electrolytic capacitors (C19, C20, C21).

LED MATRICES, LEDS, JACKS

Insert LEDs (LED1, LED2), matrices (MATRIX1, MATRIX2), potentiometer and the 16 pin (SMD male/7mm tall female) header, don't solder them yet. Orientation of various matrices:

- **LTP-747KR** text should face to right
- **TA07-11SURKWA** text should face to left
- **KWM-20571CSB** text should face to left

Slide front panel onto the pots, finger tighten nuts. Make sure the SMD header is standing on it's place. Align the matrices and LEDs, that they line up with the holes on the panel as accurately as possible. Now gingerly solder the prepared components, before moving to another leg/pad re-check alignment. Remove the panel, pay attention to the 16 pin header, pull it apart gently, not to rip off the pads and traces from the back of the panel that are already fixed. Solder the remaining pins of the SMD connector. Insert the encoder to it's place. To line up the encoder with the pots mount an extra nut on its thread. This will narrow the gap between the encoder and panel. Solder it. Lastly add the power cables, wire up banana connectors.

UPLOAD THE FIRMWARE

Cut the trace on the bottom of Teensy next to VIN pin to separate VIN from VUSB. This prevents powering the Teensy simultaneously from USB and the eurorack power supply. Install the Teensy loader application, v1.33 or later is required (www.pjrc.com/teensy/loader.html), download the hex file from www.doboz.audio. Open the hex file in the Teensy loader, connect the Teensy with a USB cable to your computer and power up the module. Press the pushbutton on the Teensy board, click the program icon (or choose program from the operation menu), click the reboot icon (or choose reboot from the operation menu). Your module should now come to life, if newly built proceed to calibration (detailed in the User's Guide).

EPILOGUE

Before inserting the Teensy, it'll make sense to check voltages indicated in the schematic. MPR121 breakout boards differ from each other. If despite proper calibration the module still behaves strangely get another MPR121 board from a different supplier. In our experience blue boards perform best. Black ones are usually alright too, however some of them required to set ***SENSITIVITY*** lower and ***DEBOUNCE*** higher (see User's Guide).
Different LED matrices may also work correctly, but before ordering carefully compare their pinout with the ones mentioned in the BOM (double check datasheets).